

Functional Requirements Specification (FRS)

**D-11 Governance by Automation
(Automated governance solution)**

March 2026



1. Introduction

1.1 Scope

This Functional Requirements Specification (FRS) defines the complete scope for Deliverable D11 – Governance by Automation, covering project governance, contract governance, meeting governance, and Retrieval-Augmented Generation (RAG)–based AI-assisted knowledge access. All AI capabilities within the solution shall operate strictly under a human-in-the-loop governance model to ensure oversight, auditability, and compliance; autonomous, unsupervised, or self-executing AI is explicitly excluded. The governance solution shall be developed using open-source technologies for perpetual use, and the use of proprietary third-party tools is not envisaged. All interactions with the Aadhaar Genius AI module shall be conducted exclusively in English.

Deliverable D11 shall be implemented as a desktop web application hosted entirely on UIDAI infrastructure and accessed through standard browsers by authorized UIDAI internal users. Considering the target user groups—UIDAI Tech Centre, TMD, and governance teams—the desktop web portal is the intended and optimal form factor.

1.2 Phased Delivery Milestones

D11 delivery is structured into 3 phased milestones:

Milestone	Module Scope	FRS Sections Covered	Delivery Timeline
Milestone 1 (Ptm)	UIDAI PMIS + Project Timeline Module + General Capabilities	Sections 2, 6	T0 + 6 months
Milestone 2 (Ctm)	Contract Governance + Aadhaar Genius (AI Data Extraction & Global Search)	Sections 3, 5	T0 + 9 months
Milestone 3 (Mtm)	Meetings & MoM Governance + Entire Tool Integration	Sections 4	T0 + 12 months

Measurement: One time per milestone — at T0+6 (Milestone 1), T0+9 (Milestone 2), T0+12 (Milestone 3), or upon completion whichever is later.

1.3 Modules Covered

1. UIDAI PMIS (Project Management Information System)
2. Project Timeline Governance Module
3. Contract Governance Module
4. Meetings & Minutes of Meeting Governance Module
5. Aadhaar Genius – AI-Driven Data Extraction & Global Search
6. General – Cross-Module Governance Capabilities

1.4 RFP Requirements Traceability Matrix

The table below provides a consolidated traceability of every RFP clause, Corrigendum amendment, SLA parameter, and bidder query to the corresponding FRS sections and requirement IDs:

Source	Reference	Description	FRS Section(s)	FRS Requirement IDs
RFP	4.2.11 (preamble)	Automated governance solution for MSP activities, PMC deliverables, contracts; configurable, flexible, hosted on UIDAI infra	1.1	GEN-FR-11 (refer Section 6.7)
RFP	4.2.11.a.i	Project Timelines Module	2	PMIS-FR-01 to PMIS-FR-51
RFP	4.2.11.a.i.A	Baseline project plan for MSP and PMC deliverables	2.7, 2.8	PMIS-FR-16 , PMIS-FR-17 , PMIS-FR-19 , PMIS-FR-20
RFP	4.2.11.a.i.B	Breakdown tasks into sub-level activities	2.6, 2.12	PMIS-FR-13 , PMIS-FR-14 , PMIS-FR-27 , PMIS-FR-28
RFP	4.2.11.a.i.C	Allocation of divisions and SPOCs to each activity	2.10, 2.11, 6.4	PMIS-FR-24 to PMIS-FR-26 , GEN-FR-06
RFP	4.2.11.a.i.D	Track revisions, delays in any activity	2.9, 2.13	PMIS-FR-21 , PMIS-FR-22 , PMIS-FR-33 , PMIS-FR-34
RFP	4.2.11.a.i.E	Audit function for tracking changes in timelines	2.4, 2.8	PMIS-FR-07 , PMIS-FR-08 , PMIS-FR-19
RFP	4.2.11.a.i.F	Comments trail for each activity for milestone changes	6.2	GEN-FR-04
RFP	4.2.11.a.i.G	Automated email notification for pending activities	2.15, 2.17	PMIS-FR-41 to PMIS-FR-43 , PMIS-FR-49 to PMIS-FR-51

RFP	4.2.11.a.i.H	Dashboard for timelines, delays per phases/divisions/SPOCs	2.16	PMIS-FR-44 to PMIS-FR-48
RFP	4.2.11.a.ii	Contract Module	3	CON-FR-01 to CON-FR-14
RFP	4.2.11.a.ii.A	Track milestones, service levels, payments	3.2	CON-FR-01 , CON-FR-02 , CON-FR-03
RFP	4.2.11.a.ii.B	Map payment schedules with milestones/SLAs	3.3	CON-FR-04 , CON-FR-05
RFP	4.2.11.a.ii.C	Upload documents, invoice workflows	3.4, 3.7	CON-FR-06 , CON-FR-11 to CON-FR-14
RFP	4.2.11.a.ii.D	Track change requests with checklists	3.5	CON-FR-07 , CON-FR-08
RFP	4.2.11.a.ii.E	Dashboard for CR fund consumption	3.6	CON-FR-09
RFP	4.2.11.a.ii.F	Alert mechanisms for SLA/event of default	3.6	CON-FR-10
RFP	4.2.11.a.iii	Meetings Module	4	MEET-FR-01 to MEET-FR-07
RFP	4.2.11.a.iii.A	Capture meeting minutes (Steering, Migration, key meetings)	4.2, 4.3	MEET-FR-01 to MEET-FR-05
RFP	4.2.11.a.iii.B	Assign activities to stakeholders with defined timelines	4.4	MEET-FR-06 , MEET-FR-07
RFP	4.2.11.a.iii.C	Track comments and updates for assigned activities	4.3	MEET-FR-04 , MEET-FR-05
RFP	4.2.11.a.iii.D	Timeline changes with approvals and audit mechanisms	4.3	MEET-FR-04
RFP	4.2.11.a.iii.E	Activity tracking and notification	4.4	MEET-FR-06 , MEET-FR-07 ,

		mechanisms		GOV-MM-05
RFP	4.2.11.a.iii.F	Dashboard for actions completed or delayed	4.4	MEET-FR-07
RFP	4.2.11.a.iv.A.1	Document management, workflows, login	6.1	GEN-FR-01 , GEN-FR-02 , GEN-FR-03
RFP	4.2.11.a.iv.A.2	Common dashboard for all modules	6.3	GEN-FR-05
RFP	4.2.11.a.iv.A.3	User access modules	2.2, 2.3	PMIS-FR-01 to PMIS-FR-06
RFP	4.2.11.a.iv.A.4	Security measures, encryption, audits	6.7	GEN-FR-09 , GEN-FR-10
RFP	4.2.11.a.iv.A.5	Integration or upload option (Word/spreadsheets)	6.5	GEN-FR-07
RFP	4.2.11.a.iv.A.6	Notifications and alerts with email	6.6	GEN-FR-08
RFP	4.2.11.a.iv.A.7	AI-driven data extraction and global search	5	AG-FR-01 , AG-FR-02 , AG-FR-03 , AG-FR-04 , AG-FR-05 , AG-FR-06 , AG-FR-07 , AG-FR-08 , GEN-FR-13
RFP	4.2.11.a.iv.A.8	Automated alerts and notifications	6.6	GEN-FR-08
RFP	4.2.11.a.iv.A.9	Defensible audit trails	2.4, 6.7	PMIS-FR-07 , PMIS-FR-08 , GEN-FR-10
RFP	4.2.11.a.iv.A.10	Integration with existing UIDAI tools	6.5	GEN-FR-07
RFP	4.2.11.a.iv.A.11	Secured centralized contract repository	6.7	GEN-FR-16

RFP	4.2.11.a.iv.B	UIDAI infrastructure hosting, security testing	6.7, 7.4	GEN-FR-11 , NFR-05
RFP	4.2.11.a.iv.C	Knowledge transfer and training	6.8	GEN-FR-12
RFP	4.2.11.a.iv.D	Maintenance and operation	6.8, 7.9	GEN-FR-12 , NFR-11
RFP	4.2.11.a.iv.E	Scheduled maintenance	6.8, 7.2	GEN-FR-12 , NFR-03
RFP	4.2.11.a.iv.F	Lifecycle management (upgrades, bug fixes)	6.8	GEN-FR-12 , GEN-FR-14 ,
RFP	4.2.11.a.iv.G	Seamless upgrades without affecting services	6.8	GEN-FR-14
RFP	4.2.11.a.iv.H.1	Upgrade individual modules independently	6.8	GEN-FR-14
RFP	4.2.11.a.iv.H.2	Revert to original configuration on failure	6.8	GEN-FR-14
RFP	4.2.11.a.iv.H.3	Compliance with DPDP and applicable Indian laws (GDPR removed per Corrigendum 17)	6.7, 7.8	GEN-FR-15 , NFR-10
SLA	5.28.4.a (SLA 010)	Deployment and Configuration of Governance Tools	1.2	Milestone delivery targets
SLA	5.28.4.b (SLA 011)	Uptime of Governance Tools (monthly; >=99.5%)	7.1, 7.2	NFR-01 , NFR-02 , NFR-03
Corrigendum	Item 16	Open-source for perpetual use; proprietary tools not envisaged	1.1	Scope statement (open-source mandate)
Corrigendum	Item 17	GDPR removed from 4.2.11.a.iv.H.3; only DPDP retained	6.7, 7.8	GEN-FR-15 (DPDP only)
Corrigendum	Item 18	Data handover clause	—	None — intentionally

		(4.2.11.a.iv.l) deleted		excluded
Corrigendum	Item 19	Phased milestones: T0+6, T0+9, T0+12 replacing single delivery	1.2, 1.3	SLA 010 milestone targets
Query	Query 91	Open-source mandate confirmed for D11	1.1	Scope statement
Query	Query 205	Custom development timeline for open-source confirmed	1.2	Milestone structure
Query	Query 277	D11 delivery timeline adjustment confirmed	1.2	SLA 010
Query	Query 286	GDPR removal confirmed per Corrigendum 17	6.7, 7.8	GEN-FR-15

2. UIDAI PMIS – Project Management Information System

2.1 Objective

UIDAI PMIS shall provide a centralized, auditable governance platform for MSP(s) and PMC projects, supporting structured planning, baseline-driven execution, dependency management, risk tracking, SLA-based issue handling, and transparent monitoring.

2.2 User Access & Authentication

PMIS-FR-01: The system shall allow access only to authorized UIDAI users.

[PMIS-FR-01.1](#): UIDAI shall onboard users in the PMIS system.

[PMIS-FR-01.2](#): Users shall be granted access based on their role, division, SPOC assignment, and project mapping.

[PMIS-FR-01.3](#): Each access request shall be evaluated against the user's organizational role and project assignment.

PMIS-FR-02: Authentication shall use UIDAI-approved authentication mechanisms.

[PMIS-FR-02.1](#): Login shall follow UIDAI security standards.

[PMIS-FR-02.2](#): No locally managed or hard-coded credentials shall be permitted.

[PMIS-FR-02.3](#): Login policies shall support organizational security rules as defined by UIDAI.

PMIS-FR-03: User sessions shall be securely managed.

[PMIS-FR-03.1](#): Sessions shall automatically expire after a defined period of inactivity.

[PMIS-FR-03.2](#): Sessions shall be terminated upon user logout.

[PMIS-FR-03.3](#): All active sessions shall be centrally visible and revocable by UIDAI security administrators.

2.3 Authorization & Role Management

PMIS-FR-04: The system shall enforce role-based access control.

[PMIS-FR-04.1](#): A role-permission matrix shall govern access to every module.

[PMIS-FR-04.2](#): Authorization shall be verified before every user action (create, edit, approve, view).

[PMIS-FR-04.3](#): Role enforcement shall apply uniformly across PMIS, Contracts, Meetings, Dashboards, and Documents.

PMIS-FR-05: Admin shall manage all project-related configuration.

[PMIS-FR-05](#).1: An Admin console shall be provided for managing users, roles, workflows, SLA rules, and program definitions.

[PMIS-FR-05](#).2: Every configuration change shall be logged for audit purposes.

[PMIS-FR-05](#).3: Configuration changes shall support versioning and rollback for governance traceability.

[PMIS-FR-05](#).4: Admin shall be able to create, modify, and deactivate UIDAI programs that projects are mapped to.

[PMIS-FR-06](#): Users shall access only projects, tasks, tickets, and contracts assigned to them.

[PMIS-FR-06](#).1: Data visibility shall be restricted based on user assignment.

[PMIS-FR-06](#).2: Unauthorized access to unassigned records shall be prevented.

[PMIS-FR-06](#).3: Division and SPOC ownership shall be enforced across all project, task, and contract views.

2.4 Audit Logging

[PMIS-FR-07](#): All user actions shall be logged with user ID, timestamp, and action type (refer [GEN-FR-10](#)).

[PMIS-FR-07](#).1: Every create, update, delete, and approve action shall be captured.

[PMIS-FR-07](#).2: Logs shall record user identity, timestamp, and the affected record.

[PMIS-FR-07](#).3: Audit logs shall also record previous and new values for governance review.

[PMIS-FR-08](#): Audit logs shall be immutable, searchable, and exportable.

[PMIS-FR-08](#).1: Logs shall be stored in tamper-proof storage.

[PMIS-FR-08](#).2: An admin interface shall be provided to search and export logs.

[PMIS-FR-08](#).3: Logs shall be retained as per UIDAI retention and regulatory compliance policies.

2.5 Project Creation & Lifecycle Management

[PMIS-FR-09](#): Authorized users shall create projects.

[PMIS-FR-09](#).1: A project creation form with validation shall be provided.

[PMIS-FR-09](#).2: Project creation shall be restricted to defined roles only.

[PMIS-FR-09](#).3: Each project shall be traceable to the approving authority through audit trails.

[PMIS-FR-10](#): Each project shall be mapped to a UIDAI program.

[PMIS-FR-10](#).1: Program selection shall be mandatory during project creation.

[PMIS-FR-10.2](#): A project cannot be saved without program mapping.

[PMIS-FR-10.3](#): Program-level roll-up reporting shall be supported for governance dashboards.

[PMIS-FR-10.4](#): UIDAI programs shall be configurable through the Admin console (refer [PMIS-FR-05.4](#)) including program name, description, owning division, and active status.

[PMIS-FR-11](#): Projects shall capture start date, end date, planned duration along with basic project metadata.

[PMIS-FR-11.1](#): Mandatory fields shall include owner, objective, planned start date, planned end date, revised start date, and revised end date.

[PMIS-FR-11.2](#): Timelines shall be validated for logical consistency (end date after start date, revised dates justified).

[PMIS-FR-11.3](#): These dates shall serve as the baseline for milestone, SLA, and contract linkage.

[PMIS-FR-11.4](#): Variance between planned and revised dates shall be automatically calculated and displayed.

[PMIS-FR-12](#): Projects shall be configurable into Activity / Tasks and sub-tasks.

[PMIS-FR-12.1](#): Admin shall be able to define standard stage templates.

[PMIS-FR-12.2](#): Stage assignment shall be required before task creation.

[PMIS-FR-12.3](#): Stage templates shall align with UIDAI PMC, MSP, and migration lifecycle governance.

2.6 Work Breakdown Structure (WBS)

[PMIS-FR-13](#): The system shall support a hierarchical Work Breakdown Structure.

[PMIS-FR-13.1](#): The WBS shall allow multi-level decomposition of work.

[PMIS-FR-13.2](#): Each level of WBS shall be independently trackable for schedule, ownership, and performance.

[PMIS-FR-13.3](#): WBS shall be the authoritative structure for governance, reporting, and contractual linkage.

[PMIS-FR-14](#): WBS levels shall include Program, Project, Sub-Project, Task, and Sub-Task.

[PMIS-FR-14.1](#): Hierarchy shall be configurable by UIDAI admin.

[PMIS-FR-14.2](#): Each level shall support assignment of UIDAI division and SPOC.

[PMIS-FR-14.3](#): Milestones, SLAs, and payments shall be linkable at any WBS level.

[PMIS-FR-15](#): Each WBS element shall have a unique identifier.

[PMIS-FR-15.1](#): A unique ID shall be system-generated.

[PMIS-FR-15.2](#): WBS IDs shall be immutable and used across contracts, meetings, tickets, and audit trails.

[PMIS-FR-15.3](#): Historical records shall always refer to the same WBS ID even after restructuring.

2.7 Baseline Planning

[PMIS-FR-16](#): The system shall allow creation of baseline project schedules.

[PMIS-FR-16.1](#): Baseline shall capture planned dates.

[PMIS-FR-16.2](#): Baseline shall be established before execution begins.

[PMIS-FR-16.3](#): Once a baseline is published, it cannot be altered.

[PMIS-FR-16.4](#): Baseline shall include transition timelines from incumbent to new service partners.

[PMIS-FR-17](#): Baselines shall include task-level, activity-level, and milestone-level timelines.

[PMIS-FR-17.1](#): Dates shall be stored at all levels.

[PMIS-FR-17.2](#): Baseline shall be used as the contractual and performance reference.

[PMIS-FR-17.3](#): Dashboards shall always show baseline vs actual.

[PMIS-FR-18](#): The system shall compute and display the critical path across the project schedule.

[PMIS-FR-18.1](#): The critical path shall identify the longest sequence of dependent tasks determining the minimum project duration.

[PMIS-FR-18.2](#): Any delay on critical path tasks shall automatically flag impact on overall project completion.

[PMIS-FR-18.3](#): Critical path shall be visually highlighted in Gantt chart and schedule views.

[PMIS-FR-18.4](#): Changes to task durations or dependencies shall trigger automatic recalculation of the critical path.

2.8 Baseline Approval & Locking

[PMIS-FR-19](#): Baseline schedules shall require approval by authorized users.

[PMIS-FR-19.1](#): An approval workflow shall be enforced.

[PMIS-FR-19.2](#): Approval shall be recorded with approver name, date, and comments.

[PMIS-FR-19.3](#): Only approved baselines shall be used for SLA, penalty, and payment calculations.

[PMIS-FR-20](#): Approved baselines shall be locked from further editing.

[PMIS-FR-20.1](#): Locked baselines cannot be changed.

[PMIS-FR-20.2](#): Any modification shall require a formal re-baseline request.

[PMIS-FR-20.3](#): Locked baselines shall remain available for audit and comparison.

2.9 Baseline Variance Management

[PMIS-FR-21](#): Progress tracking shall reference the approved baseline.

[PMIS-FR-21.1](#): Actual progress shall be compared to baseline.

[PMIS-FR-21.2](#): Delay and slippage shall be calculated automatically.

[PMIS-FR-21.3](#): Impact on milestones and payments shall be visible.

[PMIS-FR-22](#): Variance between baseline and actual dates shall be calculated automatically.

[PMIS-FR-22.1](#): Variance shall be shown in days and percentages.

[PMIS-FR-22.2](#): Variance shall be calculated at activity, task, milestone, and project level.

[PMIS-FR-22.3](#): Variance thresholds shall trigger alerts and governance reviews.

2.10 Task Management

[PMIS-FR-23](#): Tasks shall be created under defined sub-project (project stage).

[PMIS-FR-23.1](#): Tasks must belong to a stage.

[PMIS-FR-23.2](#): Stage-wise task governance shall align with UIDAI project lifecycle (PMC, MSP, Migration, Operations).

[PMIS-FR-23.3](#): Stage completion shall be driven by task and milestone status.

[PMIS-FR-24](#): Tasks shall capture description, assigned division, SPOC, planned start date, planned end date, revised start date, revised end date, and other metadata.

[PMIS-FR-24.1](#): Mandatory fields shall be enforced.

[PMIS-FR-24.2](#): Division and SPOC shall be the primary accountability owners for each task.

[PMIS-FR-24.3](#): Tasks shall be contract-traceable where applicable.

[PMIS-FR-25](#): Tasks shall be assignable to UIDAI users and approved service providers.

[PMIS-FR-25.1](#): Only authorized users can be assigned.

[PMIS-FR-25.2](#): Service partner assignments shall be validated against active contracts.

[PMIS-FR-25.3](#): Task ownership shall drive SLA and penalty calculations.

2.11 Service Provider & Team Management

PMIS-FR-26: The system shall support onboarding and offboarding of service providers and their team members.

[PMIS-FR-26.1](#): Service providers shall be registered with contract reference, validity period, and authorized team members.

[PMIS-FR-26.2](#): Upon contract termination or transition, the service provider and all associated team members shall be deactivated from the system.

[PMIS-FR-26.3](#): Deactivated service provider personnel shall lose access to all modules, tasks, and data immediately upon offboarding.

[PMIS-FR-26.4](#): Task reassignment shall be mandatory before a service provider or team member can be deactivated.

[PMIS-FR-26.5](#): A complete history of service provider team members (active and deactivated) shall be maintained for audit purposes.

2.12 Activity (Sub-Task) Management

PMIS-FR-27: Tasks shall be decomposed into activities.

[PMIS-FR-27.1](#): Activities must belong to a parent task.

[PMIS-FR-27.2](#): Activities shall be the lowest unit of execution tracking.

[PMIS-FR-27.3](#): Activity completion shall drive task and milestone status.

PMIS-FR-28: Activities shall be mapped to parent tasks.

[PMIS-FR-28.1](#): One-to-many mapping supported.

[PMIS-FR-28.2](#): Activity-level delays shall roll up to task and milestone level.

[PMIS-FR-28.3](#): Decomposition shall propagate schedule constraints to child items.

[PMIS-FR-28.4](#): Parent task status shall be computed from child item statuses.

PMIS-FR-29: Planned effort and expected completion dates shall be captured.

[PMIS-FR-29.1](#): Effort and dates shall be mandatory.

[PMIS-FR-29.2](#): These shall form part of baseline and SLA evaluation.

[PMIS-FR-29.3](#): Effort data shall be usable for productivity and capacity analysis.

[PMIS-FR-29.4](#): Effort logging shall be supported at daily, weekly, and monthly levels for detailed productivity tracking.

PMIS-FR-30: Actual start and completion dates shall be recorded.

[PMIS-FR-30.1](#): Actual dates must be captured.

[PMIS-FR-30.2](#): These shall be locked once approved.

[PMIS-FR-30.3](#): Actuals shall be used for SLA, delay, and payment calculations.

2.13 Progress & Delay Tracking

[PMIS-FR-31](#): Status values shall include Not Started, In Progress, Completed, and Delayed.

[PMIS-FR-31.1](#): Status values shall be standardized.

[PMIS-FR-31.2](#): Delayed status shall trigger alerts and dashboard flags.

[PMIS-FR-32](#): Percentage completion shall be recorded.

[PMIS-FR-32.1](#): Percentage must be between 0–100.

[PMIS-FR-32.2](#): Percentage shall be used for milestone progress and earned-value tracking.

[PMIS-FR-33](#): Delays shall be identified at task, activity, and milestone levels.

[PMIS-FR-33.1](#): Delay detection shall be automated.

[PMIS-FR-33.2](#): Delay shall be mapped to responsible division and SPOC.

[PMIS-FR-33.3](#): Delay impact on contract and payments shall be visible.

[PMIS-FR-33.4](#): Impact on the critical path shall be assessed and highlighted for delays on critical path tasks.

[PMIS-FR-33.5](#): Delays exceeding configurable thresholds shall trigger graduated alerts to SPOC, Division Head, and PMC.

[PMIS-FR-34](#): Delay responsibility shall be recorded.

[PMIS-FR-34.1](#): Responsible party must be selected.

[PMIS-FR-34.2](#): Responsibility shall be used for SLA breach attribution.

[PMIS-FR-34.3](#): Historical delay patterns shall be available for performance evaluation.

2.14 Ticket & SLA Management

[PMIS-FR-35](#): Users shall create project-related tickets.

[PMIS-FR-35.1](#): Tickets shall be linked to project, task, or activity.

[PMIS-FR-35.2](#): Tickets shall serve as the formal record for operational and contractual issues.

[PMIS-FR-35.3](#): Tickets shall be auditable and traceable.

[PMIS-FR-35.4](#): Parent-child ticket linking shall be supported to track dependencies and related issues.

[PMIS-FR-35](#).5: Bulk ticket operations (batch status updates, mass assignment) shall be supported for high-volume workflows.

[PMIS-FR-36](#): Ticket categories shall include Incident, Service Request, Change, and Problem.

[PMIS-FR-36](#).1: Categories shall be configurable.

[PMIS-FR-36](#).2: Each category shall trigger different SLA and approval workflows.

[PMIS-FR-36](#).3: Change tickets shall integrate with baseline and contract governance.

[PMIS-FR-37](#): Tickets shall capture priority, description, project reference, and assignee.

[PMIS-FR-37](#).1: Mandatory fields shall be enforced.

[PMIS-FR-37](#).2: Priority shall determine SLA timelines and escalation paths.

[PMIS-FR-37](#).3: Assignee shall be responsible for SLA compliance.

[PMIS-FR-37](#).4: Intelligent ticket routing shall automatically assign tickets based on skills matrix, team availability, and current workload.

[PMIS-FR-38](#): SLA response and resolution timelines shall be defined.

[PMIS-FR-38](#).1: SLA rules shall be configurable.

[PMIS-FR-38](#).2: Different SLAs shall apply for UIDAI, PMC, and MSP responsibilities.

[PMIS-FR-38](#).3: SLA rules shall align with contract clauses.

[PMIS-FR-39](#): SLA timers shall start automatically on ticket creation.

[PMIS-FR-39](#).1: Timers shall be system-generated.

[PMIS-FR-39](#).2: Timer shall pause or reset based on defined workflow states.

[PMIS-FR-39](#).3: All SLA time changes shall be audit logged.

[PMIS-FR-40](#): SLA timers shall reset on reassignment.

[PMIS-FR-40](#).1: Reassignment logic shall be enforced.

[PMIS-FR-40](#).2: Reassignments shall require reason capture.

[PMIS-FR-40](#).3: Frequent reassignments shall be visible to UIDAI for governance review.

2.15 Escalation Management

[PMIS-FR-41](#): SLA breaches shall trigger automatic escalations.

[PMIS-FR-41](#).1: Escalation shall be system-driven.

[PMIS-FR-41](#).2: Escalations shall be sent to UIDAI, PMC, and MSP leadership as configured.

[PMIS-FR-41](#).3: Escalation events shall be audit logged.

[PMIS-FR-42](#): Escalation rules shall be configurable.

[PMIS-FR-42](#).1: Rules shall be managed by admin.

[PMIS-FR-42](#).2: Rules shall support contract-specific escalation logic.

[PMIS-FR-42](#).3: Different severity thresholds shall be supported.

[PMIS-FR-43](#): Escalation actions shall be logged.

[PMIS-FR-43](#).1: Logs shall capture who was notified and when.

[PMIS-FR-43](#).2: Escalation logs shall be available for dispute resolution and audits.

[PMIS-FR-43](#).3: Escalation history shall be linked to contracts and penalties.

2.16 Dashboards & Reporting

[PMIS-FR-44](#): Project-level dashboards shall be available.

[PMIS-FR-44](#).1: Dashboards shall display status, milestones, risks, and delays.

[PMIS-FR-44](#).2: Dashboards shall provide UIDAI-wide visibility across all MSP(s) and PMC projects.

[PMIS-FR-44](#).3: Data shall be real-time and audit-backed.

[PMIS-FR-44](#).4: Gantt chart visualization with dependency tracking and critical path analysis shall be provided.

[PMIS-FR-44](#).5: Baseline vs. actual overlay view shall be available.

[PMIS-FR-44](#).6: Dashboard shall support drill-down to individual activity level.

[PMIS-FR-45](#): Milestone-level dashboards shall be available.

[PMIS-FR-45](#).1: Milestones shall be visually tracked.

[PMIS-FR-45](#).2: Planned vs actual vs baseline shall be shown for each milestone.

[PMIS-FR-45](#).3: Milestone delays shall be color-coded for governance review.

[PMIS-FR-46](#): Task completion and delay trend reports shall be generated.

[PMIS-FR-46](#).1: Trends shall be visualized.

[PMIS-FR-46](#).2: Reports shall show responsibility-wise (UIDAI, PMC, MSP) performance.

[PMIS-FR-46](#).3: Recurring delay patterns shall be identifiable.

[PMIS-FR-47](#): Ticket volume and aging reports shall be available.

[PMIS-FR-47](#).1: Ticket status shall be reported.

[PMIS-FR-47](#).2: Aging shall be SLA-aware and escalation-aware.

[PMIS-FR-47](#).3: High-risk vendors and teams shall be identifiable.

[PMIS-FR-48](#): Reports shall be exportable in PDF and Excel formats (refer [GEN-FR-01](#)).

[PMIS-FR-48](#).1: PDF and Excel export shall be supported.

[PMIS-FR-48](#).2: Exports shall include UIDAI watermarking and audit metadata.

[PMIS-FR-48](#).3: Exports shall be traceable to the requesting user.

2.17 Notifications

[PMIS-FR-49](#): Notifications shall be sent on task assignment (refer [GEN-FR-08](#)).

[PMIS-FR-49](#).1: Users shall receive notifications upon assignment.

[PMIS-FR-49](#).2: Notifications shall be sent to assigned SPOC and division.

[PMIS-FR-49](#).3: Delivery status shall be tracked.

[PMIS-FR-50](#): Notifications shall be sent prior to task deadlines.

[PMIS-FR-50](#).1: Reminders shall be automatic.

[PMIS-FR-50](#).2: Multiple reminder thresholds shall be configurable.

[PMIS-FR-50](#).3: Escalation shall follow if deadlines are breached.

[PMIS-FR-51](#): Notifications shall be sent on SLA breaches and escalations.

[PMIS-FR-51](#).1: Alerts shall be generated automatically.

[PMIS-FR-51](#).2: Alerts shall be routed to UIDAI, PMC, and MSP leadership.

[PMIS-FR-51](#).3: All alerts shall be logged for audit and dispute resolution.

3. Contract Governance Module

3.1 Objective

The Contract Governance Module shall provide UIDAI with centralized visibility and control over all MSP and PMC contracts, enabling milestone-linked payment tracking, SLA compliance monitoring, change request governance, and financial risk management. Payment execution is outside the scope of this module.

3.2 Contract Milestone & SLA Tracking

CON-FR-01: Contract milestones shall be tracked (refer [PMIS-FR-13](#), [PMIS-FR-14](#) for WBS linkage).

[CON-FR-01.1:](#) Milestones shall be captured and maintained.

[CON-FR-01.2:](#) Each milestone shall be mapped to project and WBS elements.

[CON-FR-01.3:](#) Milestone status shall be driven by verified task completion.

CON-FR-02: Contract SLAs shall be tracked (refer [PMIS-FR-38 to PMIS-FR-40](#) for SLA timers).

[CON-FR-02.1:](#) SLA values shall be stored and monitored.

[CON-FR-02.2:](#) SLAs shall be continuously evaluated using ticket and task data from PMIS.

[CON-FR-02.3:](#) SLA compliance shall be visible to UIDAI in real time.

CON-FR-03: Contract payment schedules shall be tracked (payment execution excluded).

[CON-FR-03.1:](#) Payment schedules shall be captured and maintained.

[CON-FR-03.2:](#) Each payment shall be linked to milestones and SLA compliance.

[CON-FR-03.3:](#) Payment eligibility shall be computed by the system.

3.3 Milestone & Payment Mapping

CON-FR-04: Contract milestones shall be mapped to project milestones.

[CON-FR-04.1:](#) Mapping shall be enforced.

[CON-FR-04.2:](#) No contract milestone shall exist without a project reference.

[CON-FR-04.3:](#) Delays in project milestones shall automatically reflect in contract tracking.

CON-FR-05: Payment schedules shall be mapped to milestones and SLAs.

[CON-FR-05.1:](#) Mapping shall be configurable.

[CON-FR-05.2:](#) Payment shall be blocked if required milestones or SLAs are not met.

[CON-FR-05.3:](#) UIDAI shall have visibility of pending and at-risk payments.

[CON-FR-05.4](#): SLA breaches shall compute penalty deductions on linked payments.

[CON-FR-05.5](#): Liquidated Damages (LD) shall be calculated based on task completion status and supporting document attachment.

[CON-FR-05.6](#): Mapping shall be configurable per contract type (MSP, PMC).

[CON-FR-05.7](#): Payment eligibility computation shall be audit-logged.

3.4 Contract Document Management

[CON-FR-06](#): Contract documents shall be stored in a centralized repository (refer [GEN-FR-16](#)).

[CON-FR-06.1](#): Documents shall be uploaded and organized.

[CON-FR-06.2](#): Versioning and access control shall be enforced.

[CON-FR-06.3](#): All contract references in PMIS shall link to this repository.

3.5 Change Request Management

[CON-FR-07](#): Contract change requests shall be logged.

[CON-FR-07.1](#): Change requests shall be recorded.

[CON-FR-07.2](#): Each change request shall reference impacted milestones, SLAs, and payments.

[CON-FR-07.3](#): Change requests shall be auditable.

[CON-FR-08](#): Change requests shall be validated using an approval process.

[CON-FR-08.1](#): An approval workflow shall be enforced.

[CON-FR-08.2](#): Approvals shall be role-based and time-stamped.

[CON-FR-08.3](#): No contract change shall take effect without UIDAI approval.

[CON-FR-08.4](#): Change request checklist templates shall be available per contract type.

[CON-FR-08.5](#): Impact assessment on milestones, SLAs, and payments shall be mandatory before approval.[CON-FR-08.6](#): Change request approval chain shall be configurable per contract.

3.6 Dashboards & Alerts

[CON-FR-09](#): Dashboards shall display fund utilization and change impact.

[CON-FR-09.1](#): Fund data shall be visualized.

[CON-FR-09.2](#): Change requests shall show approved vs consumed budgets.

[CON-FR-09.3](#): Financial risk shall be highlighted.

[CON-FR-09.4](#): Alerts shall be triggered when fund utilization exceeds configurable thresholds (70%, 85%, 95%).

[CON-FR-09.5](#): Drill-down from dashboard to individual change request details shall be supported.

[CON-FR-10](#): Alerts shall be generated for SLA and event-of-default risks.

[CON-FR-10.1](#): Alerts shall be system-driven.

[CON-FR-10.2](#): Alerts shall be triggered when contractual thresholds are breached.

[CON-FR-10.3](#): UIDAI shall be notified before any default situation.

[CON-FR-10.4](#): Event-of-default early warning shall be triggered when cumulative breaches approach contractual caps.

[CON-FR-10.5](#): Alert rules shall be developed in consultation with appointed SPOCs (ongoing exercise).

[CON-FR-10.6](#): Alert history shall be audit-logged and linked to contract records.

3.7 Invoice Management

[CON-FR-11](#): Invoices and supporting documents shall be uploaded.

[CON-FR-11.1](#): Upload capability shall be provided.

[CON-FR-11.2](#): Invoices shall be linked to contract milestones and tasks.

[CON-FR-11.3](#): Documents shall be versioned and audited.

[CON-FR-12](#): Workflow-based invoice review and approval shall be supported.

[CON-FR-12.1](#): A review workflow shall be configured.

[CON-FR-12.2](#): UIDAI financial and contract teams shall be part of the approval chain.

[CON-FR-12.3](#): All approvals shall be audit logged.

[CON-FR-12.4](#): Multi-level approval chains shall be configurable.

[CON-FR-12.5](#): Each approval step shall have configurable SLA timers.

[CON-FR-12.6](#): Rejection shall require mandatory reason capture.

[CON-FR-13](#): Invoice status shall be tracked.

[CON-FR-13.1](#): Status shall be visible at all times.

[CON-FR-13.2](#): Status shall include Pending, Approved, Held, Rejected, and Ready for Payment.

[CON-FR-13](#).3: Reasons for holds shall be recorded.

[CON-FR-14](#): Invoice actions shall be audit-logged.

[CON-FR-14](#).1: All actions shall be logged.

[CON-FR-14](#).2: Invoice logs shall be available for regulatory and audit scrutiny.

[CON-FR-14](#).3: Logs shall include approver, amount, and decision.

4. Meetings & Minutes of Meeting Governance Module

4.1 Objective

The Meetings & MoM Governance Module shall provide UIDAI with a structured, auditable system for capturing meeting outcomes, assigning and tracking action items, and ensuring accountability across all governance, steering, and migration meetings involving UIDAI, PMC, and MSP stakeholders.

4.2 Meeting Capture & Classification

MEET-FR-01: Project-related meetings shall be captured.

MEET-FR-01.1: Meetings shall be recorded in the system.

MEET-FR-01.2: Meetings shall be linked to projects, stages, and service providers.

MEET-FR-01.3: Meeting records shall be auditable.

MEET-FR-02: Meetings shall be classified as Steering, Governance, or Migration.

MEET-FR-02.1: Meeting type shall be selected during creation.

MEET-FR-02.2: Each type shall have its own approval and reporting workflow.

MEET-FR-02.3: UIDAI shall be able to filter and report by meeting category.

MEET-FR-02.4: Meeting types shall be configurable (Steering, Governance, Migration, Ad-hoc).

MEET-FR-03: Meeting date, participants, and agenda shall be recorded.

MEET-FR-03.1: Mandatory fields shall be enforced.

MEET-FR-03.2: Participants shall be validated against authorized roles.

MEET-FR-03.3: Agenda shall be preserved for audit and traceability.

4.3 Minutes of Meeting

MEET-FR-04: Minutes of Meeting shall be recorded.

MEET-FR-04.1: Users shall enter MoM content.

MEET-FR-04.2: MoMs shall be structured into decisions, actions, and risks.

MEET-FR-04.4: Threaded comment history shall be maintained per action item.

MEET-FR-04.5: Timeline extension requests shall require documented justification and approval by designated authority with audit logging.

MEET-FR-05: Structured MoM records shall be maintained.

[MEET-FR-05](#).1: Standardized MoM templates shall be used.

[MEET-FR-05](#).2: Each MoM shall be searchable and linked to tasks, milestones, and tickets.

[MEET-FR-05](#).3: MoMs shall form part of the official UIDAI project record.

4.4 Action Item Tracking

[MEET-FR-06](#): Action items shall be derived from MoMs (refer [PMIS-FR-23](#) for task creation).

[MEET-FR-06](#).1: Actions shall be created from meeting decisions.

[MEET-FR-06](#).2: Actions shall be assigned to UIDAI, PMC, or MSP SPOCs.

[MEET-FR-06](#).3: Actions shall be linked to PMIS tasks where applicable.

[MEET-FR-07](#): Action items shall capture owner, target date, and status.

[MEET-FR-07](#).1: Fields shall be mandatory.

[MEET-FR-07](#).2: Action delays shall be tracked and escalated.

[MEET-FR-07](#).3: Status changes shall be audit logged.

[MEET-FR-07](#).4: Action items shall create linked PMIS tasks when configured.

[MEET-FR-07](#).5: Email alerts for overdue items shall be sent to owner and escalation authority.

[MEET-FR-07](#).6: Dashboard shall show completion rate by meeting type, division, and SPOC with drill-down to specific action items.

5. Aadhaar Genius – AI-Driven Data Extraction & Global Search

5.1 Objective

Aadhaar Genius shall provide intelligent extraction of structured data from uploaded governance documents and a unified search capability across all modules of the D11 platform, as required under RFP clause 4.2.11.a.iv.A.7. All AI operations shall use UIDAI-approved data only and shall not access external sources.

Delivery: Milestone 2 (T0+9 months).

5.2 AI-Driven Data Extraction

[AG-FR-01](#): The system shall automatically extract key structured data from uploaded governance documents.

[AG-FR-01](#).1: Extraction shall support contracts, SLAs, RFP clauses, circulars, and meeting minutes.

[AG-FR-01](#).2: Extracted fields shall include milestones, dates, penalty clauses, payment terms, SLA parameters, and parties involved.

[AG-FR-01.3](#): Extracted data shall be presented to the user for review and confirmation before populating module fields.

[AG-FR-02](#): The system shall support extraction from multiple document formats.

[AG-FR-02.1](#): PDF documents shall be supported.

[AG-FR-02.2](#): Microsoft Word documents (.doc, .docx) shall be supported.

[AG-FR-02.3](#): Spreadsheet files (.xls, .xlsx) shall be supported for tabular data extraction.

[AG-FR-03](#): Extracted data shall auto-populate relevant module fields where applicable.

[AG-FR-03.1](#): Extracted milestones and dates shall be mappable to PMIS and Contract module fields.

[AG-FR-03.2](#): Extracted SLA parameters shall be mappable to SLA configuration in the Contract module.

[AG-FR-03.3](#): Auto-population shall require user confirmation; no field shall be overwritten without explicit approval.

[AG-FR-04](#): Document metadata shall be automatically tagged upon upload.

[AG-FR-04.1](#): AI shall identify document type (contract, circular, SLA, RFP, MoM) from content.

[AG-FR-04.2](#): AI shall extract and suggest metadata values (contract reference, validity period, parties, document date).

[AG-FR-04.3](#): Suggested metadata shall be editable by the user before saving.

[AG-FR-05](#): All extraction activities shall be audit-logged.

[AG-FR-05.1](#): Each extraction event shall log the source document, extracted fields, user who confirmed, and timestamp.

[AG-FR-05.2](#): Extraction confidence scores shall be recorded for governance review.

[AG-FR-05.3](#): Rejected or corrected extractions shall be logged with the user's correction.

5.3 Global Search

[AG-FR-06](#): A unified global search shall be provided across all D11 modules.

[AG-FR-06.1](#): A single search interface shall query PMIS (projects, tasks, tickets), Contracts (milestones, payments, change requests), Meetings (MoMs, action items), and Documents simultaneously.

[AG-FR-06.2](#): Search results shall be grouped by module with clear source indication.

[AG-FR-06.3](#): Search shall cover both structured data (fields, metadata) and unstructured content (document text, comments, MoM content).

AG-FR-07: Search results shall be AI-ranked by relevance.

[AG-FR-07.1](#): Results shall be ranked using AI/NLP-based relevance scoring, not just keyword matching.

[AG-FR-07.2](#): Search within documents shall operate at clause and section level, returning specific paragraphs rather than entire files.

[AG-FR-07.3](#): All AI-generated search rankings shall be traceable to source data.

AG-FR-08: Search results shall respect role-based access and support filtering.

[AG-FR-08.1](#): Users shall only see results from records and documents they are authorized to access.

[AG-FR-08.2](#): Filters shall be provided for module, document type, date range, division, SPOC, project, and contract.

[AG-FR-08.3](#): All search queries shall be audit-logged.

6. General – Cross-Module Governance

6.1 Document Management

GEN-FR-01: Centralized document management shall be provided across modules.

[GEN-FR-01.1](#): All documents shall be stored in a single centralized repository.

[GEN-FR-01.2](#): Documents shall be linked to projects, contracts, milestones, MoMs, and tickets.

[GEN-FR-01.3](#): UIDAI shall have complete visibility and control over all documents.

GEN-FR-02: Document versioning and metadata shall be maintained.

[GEN-FR-02.1](#): All document versions shall be preserved.

[GEN-FR-02.2](#): Metadata shall include owner, project, date, document type, and status.

[GEN-FR-02.3](#): No document shall be overwritten without retaining its history.

GEN-FR-03: Role-based access shall be enforced for documents.

[GEN-FR-03.1](#): Access permissions shall be applied to all documents.

[GEN-FR-03.2](#): Sensitive documents (contracts, invoices, audit files) shall be restricted to authorized users.

[GEN-FR-03.3](#): All document access shall be audit logged.

6.2 Comments & Collaboration

GEN-FR-04: Persistent comment trails shall be maintained for timelines.

[GEN-FR-04](#).1: Comments shall be stored against timeline items.

[GEN-FR-04](#).2: Comments shall be preserved across baseline changes.

[GEN-FR-04](#).3: Comments shall be visible during audit and dispute resolution.

6.3 Unified Dashboard

[GEN-FR-05](#): Unified governance dashboard shall consolidate all modules.

[GEN-FR-05](#).1: All modules shall feed into one consolidated dashboard.

[GEN-FR-05](#).2: UIDAI shall see projects, contracts, meetings, SLAs, and payments in one view.

[GEN-FR-05](#).3: Dashboards shall reflect real-time data and respect user role permissions.

[GEN-FR-05](#).4: Multi-dimensional filter support shall be provided (by service partner, project, module, time period, SLA metric, financial metric, and risk level) with AND/OR logic.

[GEN-FR-05](#).5: One-click drill-down from aggregate KPIs to detail level shall be supported.

[GEN-FR-05](#).6: Customizable dashboard layouts per user role (Minister, CEO, DDG, Division Head) shall be supported.

6.4 Division & SPOC Management

[GEN-FR-06](#): Division and SPOC allocation shall be supported at all levels.

[GEN-FR-06](#).1: Allocation shall be configurable by UIDAI admin.

[GEN-FR-06](#).2: Every project, task, contract, MoM, and ticket shall have an assigned SPOC and division.

[GEN-FR-06](#).3: SPOC performance shall be measurable across all modules.

6.5 Integration

[GEN-FR-07](#): Secure integration with UIDAI systems shall be supported.

[GEN-FR-07](#).1: System interfaces shall be used for inter-system data exchange.

[GEN-FR-07](#).2: Integration shall follow UIDAI security and data exchange standards.

[GEN-FR-07](#).3: File upload option for Word and spreadsheet formats shall be supported for integration with existing audit and SLA management tools.

[GEN-FR-07](#).4: All data exchange shall be logged.

6.6 Notifications

[GEN-FR-08](#): Centralized notification management shall be available.

[GEN-FR-08.1](#): Notifications shall be controlled centrally across all modules.

[GEN-FR-08.2](#): UIDAI shall configure alert thresholds and escalation paths.

[GEN-FR-08.3](#): Email and in-app alerts shall be supported.

6.7 Security & Compliance

[GEN-FR-09](#): Security controls and audits shall be enforced.

[GEN-FR-09.1](#): Data shall be protected in transit and at rest.

[GEN-FR-09.2](#): Security standards defined by UIDAI shall be followed.

[GEN-FR-09.3](#): Regular security audits shall be supported.

[GEN-FR-10](#): Defensible audit trails shall be maintained.

[GEN-FR-10.1](#): Audit logs shall be maintained across all modules.

[GEN-FR-10.2](#): All actions shall be traceable to a user, timestamp, and record.

[GEN-FR-10.3](#): Audit data shall support legal and regulatory scrutiny.

[GEN-FR-11](#): Hosting shall be on UIDAI-owned infrastructure.

[GEN-FR-11.1](#): The solution shall be deployed on UIDAI infrastructure.

[GEN-FR-11.2](#): No UIDAI production data shall be hosted externally.

[GEN-FR-11.3](#): UIDAI shall control all servers, storage, and access.

6.8 Upgrades, Rollback & Operations

[GEN-FR-12](#): The system shall support upgrades and rollback capabilities.

[GEN-FR-12.1](#): Upgrades shall be performed with minimal disruption to users.

[GEN-FR-12.2](#): Rollback shall be possible in case of failure.

[GEN-FR-13](#): AI-driven global search and data extraction across modules shall be provided using UIDAI-approved data only.

[GEN-FR-13.1](#): AI-assisted search shall be supported across all modules.

[GEN-FR-13.2](#): AI shall retrieve and summarize only UIDAI-authorized content.

[GEN-FR-13.3](#): All AI responses shall be traceable to source documents.

[GEN-FR-14](#): Module upgrades shall be performed with minimal planned downtime.

[GEN-FR-14.1](#): Individual modules shall be upgradable independently.

[GEN-FR-14.2](#): Rollback to the original configuration shall be supported in case of upgrade failure.

[GEN-FR-14.3](#): Upgrade history shall be audit logged.

[GEN-FR-14.4](#): Planned maintenance windows shall be communicated to UIDAI in advance; best efforts shall be made to minimize service disruption.

[GEN-FR-15](#): The solution shall comply with DPDP and other applicable Indian laws.

[GEN-FR-15.1](#): Data protection rules shall be enforced at all levels.

[GEN-FR-15.2](#): Personal data handling shall follow DPDP requirements.

[GEN-FR-15.3](#): Compliance status shall be reviewable by UIDAI.

[GEN-FR-16](#): A secured centralized contract repository shall be maintained.

[GEN-FR-16.1](#): All contract documents, amendments, and change requests shall be stored centrally.

[GEN-FR-16.2](#): Access control shall be enforced per role and contract sensitivity.

[GEN-FR-16.3](#): Repository shall be searchable by contract, vendor, date, and type.

7. Non-Functional Requirements (NFR)

7.1 Performance

[NFR-01](#): The system shall respond to user interactions within acceptable time limits.

[NFR-01.1](#): Standard pages and dashboards shall load within 3 seconds under normal load.

[NFR-01.2](#): Report generation (PDF/Excel) shall complete within 15 seconds for standard reports.

[NFR-01.3](#): Aadhaar Genius search and extraction responses shall be generated within 10 seconds for typical queries.

[NFR-02](#): The system shall support the expected concurrent user load.

[NFR-02.1](#): The system shall support a minimum of 100 concurrent users without performance degradation.

[NFR-02.2](#): Peak load handling shall be designed for up to 200 concurrent users.

7.2 Availability

[NFR-03](#): The system shall maintain high availability during business hours.

[NFR-03.1](#): Target availability shall be 99.5% measured monthly, excluding planned maintenance windows.

NFR-03.2: Planned maintenance shall be scheduled during non-business hours with prior notice to UIDAI.

NFR-03.3: Unplanned downtime incidents shall be documented and reported to UIDAI.

7.3 Scalability

NFR-04: The system shall be scalable to accommodate growth in data, users, and documents.

NFR-04.1: The system shall handle growing volumes of projects, contracts, tickets, and meeting records without performance degradation.

NFR-04.2: The Aadhaar Genius document corpus shall be scalable to accommodate thousands of documents.

NFR-04.3: The system architecture shall allow horizontal scaling of components as usage grows.

7.4 Security

NFR-05: The system shall follow UIDAI security policies and industry best practices (refer GEN-FR-09, GEN-FR-11).

NFR-05.1: All data shall be protected in transit and at rest.

NFR-05.2: Vulnerability assessment and penetration testing shall be completed before go-live and periodically thereafter.

NFR-05.3: Security patches shall be applied within timelines agreed with UIDAI.

NFR-05.4: Access to production systems shall be restricted to authorized personnel only.

7.5 Backup & Recovery

NFR-06: Data backup and recovery procedures shall be established.

NFR-06.1: Regular automated backups shall be performed as per UIDAI backup policy.

NFR-06.2: Recovery Point Objective (RPO) shall be no more than 24 hours.

NFR-06.3: Recovery Time Objective (RTO) shall be no more than 4 hours for critical modules.

NFR-06.4: Backup restoration shall be tested periodically.

7.6 Usability

NFR-07: The system shall be usable by UIDAI personnel with standard computer skills.

NFR-07.1: The user interface shall be intuitive and consistent across all modules.

NFR-07.2: Online help and contextual guidance shall be available within the application.

7.7 Compatibility

NFR-08: The system shall be accessible through standard web browsers.

NFR-08.1: The application shall support the latest versions of Chrome, Firefox, and Edge browsers.

NFR-08.2: The interface shall be responsive and functional on standard desktop resolutions.

7.8 Data Retention & Archival

NFR-09: Data retention shall comply with UIDAI policies.

NFR-09.1: Active data shall be retained for the duration of the project and contract lifecycle.

NFR-09.2: Archived data shall be accessible for audit and regulatory purposes.

NFR-09.3: Data purging shall be performed only with UIDAI authorization.

7.9 Maintainability

NFR-10: The solution shall be maintainable and supportable by UIDAI or its designated partners.

NFR-10.1: Source code shall be well-documented and follow coding standards.

NFR-10.2: Configuration shall be externalized and manageable without code changes.

NFR-10.3: Logging shall be comprehensive for troubleshooting and monitoring.